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1. Introduction

Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs). Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [1]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones, or it can be used to examine individual vessels with measurements of diameter or tortuosity. Nevertheless, OCT is prone to certain artifacts which specifically affect visualization of the vasculature. These artifacts may lead to underestimations of measurements of vessel diameter and density. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels. To compensate for these artifacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished. Here we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. We finally propose ways of mitigating or correcting for underestimations of vessel diameter related to typical OCT artifacts.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample. OCT has found widespread clinical use in ophthalmology with XXX number of systems and XXX OCT scans performed annually around the world. OCT angiography (OCTA) is a functional extension of OCT which highlights motion between repeated OCT scans to generate images of vascular networks. Typically, OCTA signals are first calculated from the OCT intensity and/or phase information, segmented in 3D manually, algorithmically, or using a deep learning approach, and finally compressed into a 2D image using a maximum intensity projection. These 2D vascular maps can then be binarised by using a global or adaptive threshold, and finally vascular metrics

such as vessel density and diameter can be computed. Of these processing stages, the thresholding step is perhaps the most important. There are dozens of different thresholding methods that have been proposed over the years and each of these may produce different results in the binary map, which would in turn influence the quantification of vascular metrics. In this work, we examine an alternate approach for extracting quantitative vessel diameter by fitting a model to the OCTA data before binarisation to avoid thresholding-based effects.

In addition to thresholding-based variability, we hypothesize that OCTA metrics are also affected by OCT artifacts caused by the non-uniform backscattering profiles of red blood cells (RBCs). Because RBCs have a distinct bi-concave disc shape, they more effectively backscatter light when aligned perpendicular to the beam, rather than parallel to it. It has already been observed that in large vessels, a so-called hourglass artifact appears when viewed cross-sectionally. This is caused by the alignment of the RBCs with the vessel walls while under laminar flow, where the RBCs are perpendicular to the beam in the center of the vessels and parallel to the beam at the edges of the vessels. This artifact artificially suppresses the OCT and OCTA signals at the edges of the vessels, making them appear more narrow when projected into an en-face view, which is the preferred way of displaying angiographic data. Here we use contrast enhanced OCT (CE-OCT) to investigate how significant these effects are and how they can be corrected.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [2]. The system was centred at 840 nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 μm in air and 3.8 μm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at an A-scan rate of 80 kHz and a beam diameter of 0.5 mm at the pupil plane (power at cornea = 2.85 mW) [3]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a custom 502 nm light-emitting diode (LED) ring flashing at 10 Hz (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~500 nm), enabling predominantly rod-driven photoreceptor stimulation [4].

2.2. Experimental protocol

Three mouse models were used in this study: VLDLR knockout mice (B6;129S7-*Vldlr*^{tm1Her}/J, The Jackson Laboratory, Bar Harbor, USA), C57BL/6 mice (C57BL/6, The Jackson Laboratory, Bar Harbor, USA), and SOD1 transgenic mice (strain details, The Jackson Laboratory, Bar Harbor, USA), with $n = \text{XX}$ mice per group. All imaging sessions were conducted under general anaesthesia using either isoflurane or a ketamine/xylazine combination. Isoflurane was induced at 4% in oxygen for 4 minutes and subsequently maintained at 2% in oxygen at a flow rate of 1.5–2 L/min. In cases where isoflurane alone did not provide adequate sedation, an intraperitoneal injection of ketamine/xylazine cocktail (100 mg/kg ketamine, 6 mg/kg xylazine; 10 mL/kg body weight) was administered. To maintain core body temperature of the mice throughout imaging, they were positioned on a heating pad. Mouse pupils were dilated with one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at regular intervals to prevent the cornea from drying out.

Intralipid has been shown to be a highly scattering contrast agent that fills the vascular lumen uniformly, thereby compensating for the artefact caused by the diminished backscattering at vessel peripheries caused by RBC orientation effects [3,5,6]. For this study, Intralipid 20% (3 mL/kg

body weight) was administered as a bolus injection via the tail vein. Volumetric angiography data were acquired over a 1 mm × 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). For this study, pre- and post-injection datasets and pre- and post-stimulation datasets were acquired. **These volumetric data were used solely for qualitative visualisation of the contrast enhancement effect across the retinal vascular plexuses and were not used for quantitative diameter analysis.** Additionally, a dynamic-contrast OCT (DyC-OCT) protocol [7] was employed during the contrast injection or during stimulation, consisting of 2000 repeated B-scans over approximately 16 seconds (interscan time 8.2 ms) at the same cross-sectional location over a 1 mm lateral field of view. DyC-OCT scans allow us to track the same vessels over time to precisely measure the effects of the contrast agent or the stimulation with a high degree of spatial coregistration. **All quantitative vessel diameter measurements reported in this study were derived from the DyC-OCT data.**

Functional flicker stimulation was performed in a C57BL/6 mouse to assess whether the choice of diameter measurement method affects the detection of physiological vascular responses. For this experiment, ketamine/xylazine cocktail was chosen over isoflurane to minimise anaesthesia-induced vasodilation. A DyC-OCT acquisition was performed during which the retina was unstimulated for approximately 4 seconds (baseline), followed by 10 Hz flicker stimulation from the LED ring for approximately 7.5 seconds, and a post-stimulation period of approximately 4.5 seconds **(I wrote this to match the pre- and post-stim lines in the stimulation result plots) CM[This isn't the stimulation protocol that was used though. The lines just mark where we were averaging, but I believe in this data set there was already stimulation before the data started acquiring and it continued through the full acquisition. Do you remember how much this was?].**

All animal procedures were approved by the ethics committee of the Medical University of Vienna and the Austrian Federal Ministry of Education, Science, and Research (BMBWF/66.009/0272-V/3b/2019 and GZ 2024-0.802.524).

2.3. Post-processing and analysis

Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) were used for inter-frame alignment and B-scan flattening [3, 8]. Angiograms were derived by applying the complex subtraction OCT angiography (OCTA) algorithm [9, 10]. We evaluated multiple combinations of methods for quantifying vessel diameter from the OCTA data to determine which combinations of steps work best under different conditions. Specifically, we looked at two types of data projections (maximum intensity and mean intensity) to generate vessel profiles and several different quantifications of those profiles including a traditional binarisation, a Gaussian model approach, and a Generalised-Gaussian (GG) model approach. These methods represent commonly used methods in the literature for OCTA quantification [11–14]. The different methods were evaluated against the ground truth as described in this section. All data processing and analysis was performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

2.3.1. Ground truth calculation

Ground truth vessel diameters were obtained from contrast-enhanced DyC-OCT B-scan angiograms, where the uniformly scattering Intralipid fills the full vascular lumen regardless of RBC orientation. The width and the height of the vessel were manually measured from the vessel's cross-section. The ground truth diameter was defined as the minimum of the two measurements to account for off-axis cross-sectioning of the vessel, which produces a more elliptical appearance in the B-scan. For cases where the vessel is perpendicular to the B-scan, the height and width would therefore be in agreement. Because the lateral diameter is studied here, we defined a correction factor of *height/width* for cases where the width was larger than the height due to skew effects and was equal to 1 otherwise. By multiplying this correction factor against diameter

141 measured from the ROI projection, we ensured that the measured vessel diameter reflected the
 142 true size of the vessel.

143 2.3.2. Vessel projection calculation

144 For each vessel of interest, a rectangular region of interest (ROI) encompassing the vessel cross-
 145 section, along the lateral (x) and axial (z) dimensions, was selected from the DyC-OCT B-scan
 146 angiograms. En face intensity profiles were then generated by projecting the A-scans within this
 147 ROI along the axial (z) direction yielding a time-stack of 1-D lateral intensity profiles for each
 148 vessel. Two projection methods were compared: maximum intensity projection (MIP), which
 149 selects the highest intensity value along the depth axis of each ROI, and mean intensity projection
 150 (MeIP), which averages the values along each ROI. Each projection resulted in a time-stack of
 151 one-dimensional intensity profiles for each vessel, which were then used for subsequent diameter
 152 quantification. For the model-based approaches, the profiles were normalised to a range of 0–1
 153 to ensure a good fit.

154 2.3.3. Model-based vessel diameter calculation

155 The two parametric models were fitted (Fig. 1a) to the one-dimensional intensity profiles from
 156 the previous step, using nonlinear least-squares regression in MATLAB.

157 Standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) + C \quad (1)$$

158 where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a
 159 constant baseline offset.

160 Generalised-Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x - \mu|}{\alpha}\right)^\beta\right) + C \quad (2)$$

161 where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape
 162 parameter, and C is a constant baseline offset. Given the imaging system configuration, all vessels
 163 are expected to produce at minimum a Gaussian intensity profile in the B-scan cross-section,
 164 and so profiles sharper than Gaussian profile ($\beta < 2$) are therefore not technically meaningful.
 165 The shape parameter β was hence constrained to $\beta \geq 2$ during fitting. With this constraint, the
 166 GG accommodated to profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$) to increasingly
 167 flat-topped ($\beta > 2$).

168 Two standard width metrics were calculated from the fitted models: the full width at half
 169 maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was
 170 calculated as:

$$w_{1/e^2, G} = 4\sigma \quad (3)$$

171 The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$\text{FWHM}_G = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2} w_{1/e^2, G} \quad (4)$$

172 and therefore only the $1/e^2$ width was calculated directly from the fit, with the FWHM obtainable
 173 via this known relationship.

174 For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since
 175 both depend on the shape parameter β . Both metrics were therefore calculated independently as:

$$\text{FWHM}_{GG} = 2\alpha(\ln 2)^{1/\beta} \quad (5)$$

176

$$w_{1/e^2, GG} = 2\alpha \cdot 2^{1/\beta} \quad (6)$$

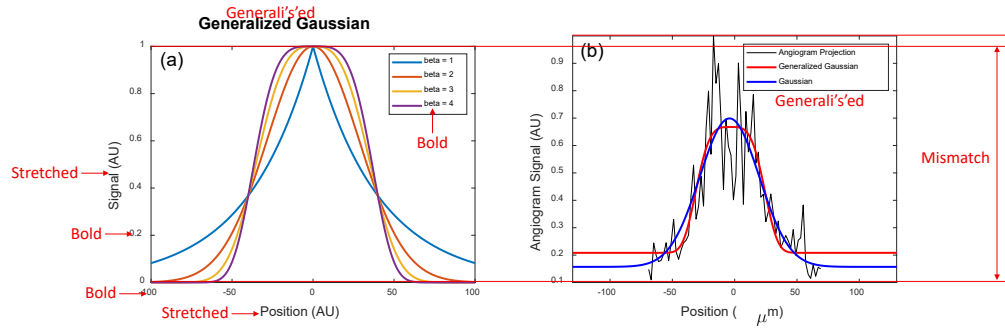


Fig. 1. (a) GG function for shape parameters $\beta = 1, 2, 3$, and 4 . At $\beta = 2$ the function reduces to a standard Gaussian; higher β values produce increasingly flat-topped profiles. (b) Example en face vessel intensity profile (black) with standard Gaussian (blue) and GG (red) fits.

2.3.4. Binary mask based vessel diameter calculation

For comparison with OCTA processing pipelines that rely on image binarisation, vessel diameters were also measured from binary masks generated by global thresholding. A 3D averaging filter with a kernel size of $50 \times 50 \times 10$ voxels (lateral $x \times$ axial $z \times$ temporal t) was applied to the DyC-OCT angiogram volumes to reduce noise and minimize the effects of outliers. To produce a binarised black and white (BW) mask, the maximum angiogram intensity within these smoothed volumes were calculated and five different threshold values were defined as fixed percentages of these maxima: **BW-1 = XX%, BW-2 = XX%, BW-3 = XX%, BW-4 = XX%, and BW-5 = XX%**. Each threshold was applied to the unsmoothed en face projection (MIP and MeIP) to generate a binary mask, and the vessel diameter was measured as the number of pixels above the threshold across the lateral vessel cross-section.

2.3.5. Statistical analysis

To evaluate the accuracy and precision of all diameter measurement approaches, the calculated diameter values were plotted against the ground truth for each vessel across three conditions: pre-contrast, peak contrast, and post-contrast. A linear regression was fitted to each condition, yielding the slope (m) and coefficient of determination (R^2). A slope of $m = 1$ indicated perfect agreement with the ground truth, and R^2 quantified the goodness of fit of the linear model. Additionally, the coefficient of variation (CoV) was computed for each method to assess measurement precision (repeatability). For the model-based methods, six combinations were evaluated: two projection methods (MIP and MeIP) by three width metrics ($w_{1/e^2, G}$, $FWHM_{GG}$, and $w_{1/e^2, GG}$). For the binarisation-based methods, diameter was measured at five threshold levels (BW-1 through BW-5) for both MIP and MeIP. For the functional flicker stimulation experiment, the Pearson correlation coefficient (ρ) between measured diameter values and angiogram intensity was computed for each method and vessel, to assess whether the diameter measurements were affected by changes in angiogram intensity. Measurements with $p < 0.05$ were considered statistically significant.

3. Results

3.1. Qualitative assessment of vessel underestimation

Figure 2 shows OCTA MIP enface images of the superficial vascular plexus (SVP), intermediate capillary plexus (ICP), and deep capillary plexus (DCP) before and after contrast injection. Across

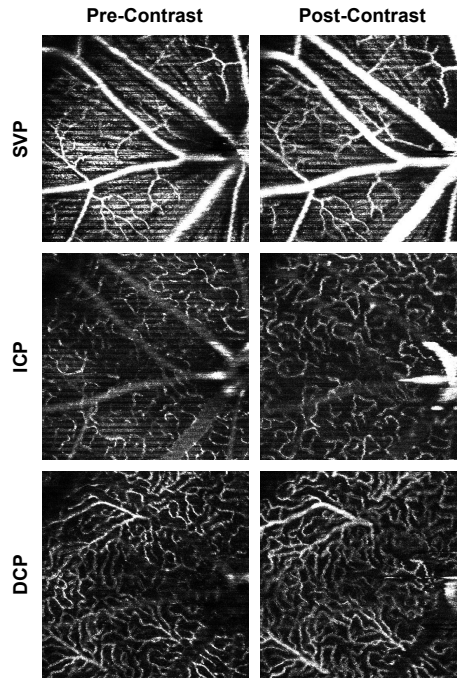


Fig. 2. OCTA MIP en face of the SVP, ICP, and DCP before (left) and after (right) the administration of Intralipid 20%. Vessels appear visibly wider and brighter post-contrast across all layers, most prominently in the larger vessels.

all three layers, vessels appeared visibly wider and brighter post-contrast injection, consistent with previous literature showing that OCTA underestimates vessel diameter in the absence of a contrast agent [3]. This effect was most pronounced in the larger vessels, especially in the SVP, where the major retinal arteries and veins appeared noticeably thicker after contrast administration. In the ICP and DCP, the capillary networks also showed improved visibility, though the effect was less dramatic.

DyC OCTA projections helped visualise the temporal changes in vessel diameter pre- and post-contrast injection (Fig. 3). Pre-contrast injection, the vessel cross-sections appeared narrower than their true diameter (Fig. 3a), which could be due to lateral edge signal suppression by the RBC orientation artefact [15]. At peak contrast, the Intralipid filled the vascular lumen uniformly and showed the full extent of the vessel cross-section (Fig. 3b). However, the high scattering nature of the contrast agent produced light diffusion beyond the vessel wall, resulting in an apparent vessel diameter that slightly exceeded the true anatomical diameter. In the post-contrast phase, the signal stabilised as the initial bolus cleared, and the vessel boundaries remained more visible than before injection without the overestimation observed at peak contrast (Fig. 3c). The MIP and the corresponding binary mask over time (Figs. 3d–e) showed the observed temporal changes in vessel diameter across the three phases. The mean OCTA intensity (Fig. 3f) showed a sharp increase post-contrast injection, followed by a decrease to a new steady state that remained elevated above the pre-contrast baseline. These observations confirm that the contrast agent does significantly enhance vessel visibility and diameter, and that the peak- and post-contrast images show improved vessel delineation compared to pre-contrast.

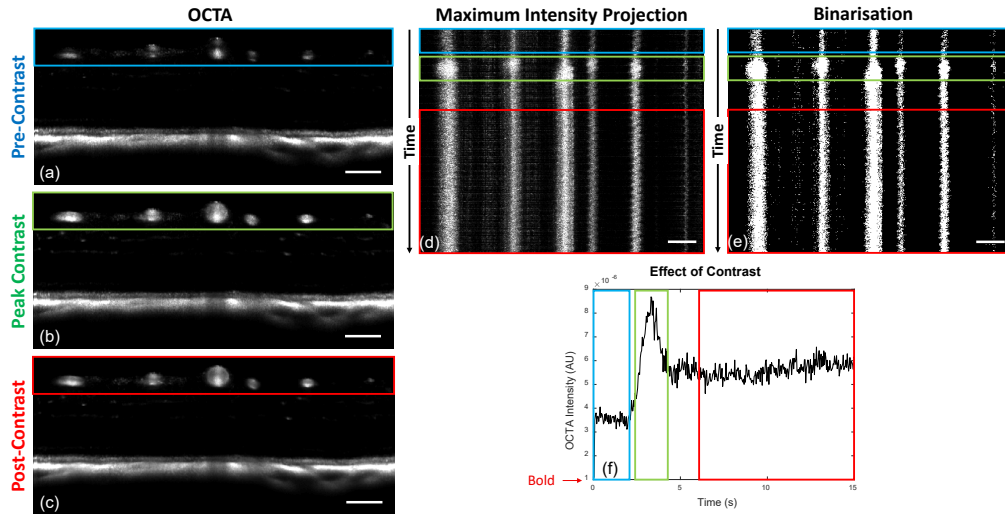


Fig. 3. DyC-OCT B-scan angiograms at (a) pre-contrast, (b) peak contrast, and (c) post-contrast timepoints. (d) MIP and (e) binarisation of the vessel cross-sections over time, with coloured boxes indicating the pre-contrast (blue), peak-contrast (green), and post-contrast (red) windows. (f) Mean OCTA intensity over time, showing the temporal intensity dynamics. All scale bars are 100 μm .

3.2. Accuracy and precision of diameter measurements

The accuracy (m , R^2) and precision (CoV) of all diameter measurement methods are summarised in Table 1.

All model-based methods underestimated vessel diameter in pre-contrast data, with slopes ranging from 0.42 to 0.73 (Table 1). Contrast enhancement improved accuracy across all metrics, though the degree of improvement varied. The FWHM_{GG} metric remained well below unity even post-contrast. The $w_{1/e^2, \text{G}}$ with MIP slightly overestimated post-contrast ($m = 1.12\text{--}1.15$), while MeIP yielded values closer to unity ($m = 1.04$). The $w_{1/e^2, \text{GG}}$ with MeIP provided the best overall agreement, achieving a slope closest to unity with $m = 1.001$ ($R^2 = 0.97$) post-contrast, which was the closest to the identity line of all tested model-based methods. Across all metrics, MeIP mostly outperformed MIP in both accuracy and linearity (Table 1, Fig. 4).

For binary mask based methods, the slope decreased with increasing threshold across all contrast conditions and projections (Table 1). This degradation was steeper for MIP than for MeIP. MIP slopes dropped from near unity at BW-1 to well below it at BW-5, whereas MeIP slopes declined more gradually and remained closer to unity across the threshold range. MeIP slopes were consistently higher than MIP at every threshold and contrast condition, though at lower thresholds and at peak and post-contrast, MeIP tended to slightly overestimate vessel diameter. The R^2 values followed a similar pattern. For MIP, pre- and post-contrast R^2 decreased substantially with increasing threshold, whereas for MeIP, R^2 remained relatively stable. Overall, MeIP provided more reliable binarisation-based diameter measurements than MIP.

The precision of each method was assessed via the CoV (Table 1, Fig. 7). For MIP, the binary mask based CoV increased substantially with threshold, whereas the model-based CoV remained stable across all contrast conditions. For MeIP, the binary mask based CoV was notably more stable across thresholds than for MIP, and the model-based CoV values were lower overall, though they increased slightly at peak contrast before decreasing post-contrast. Across both methods, MeIP yielded lower CoV values than MIP pre-contrast and more stable values post-contrast.

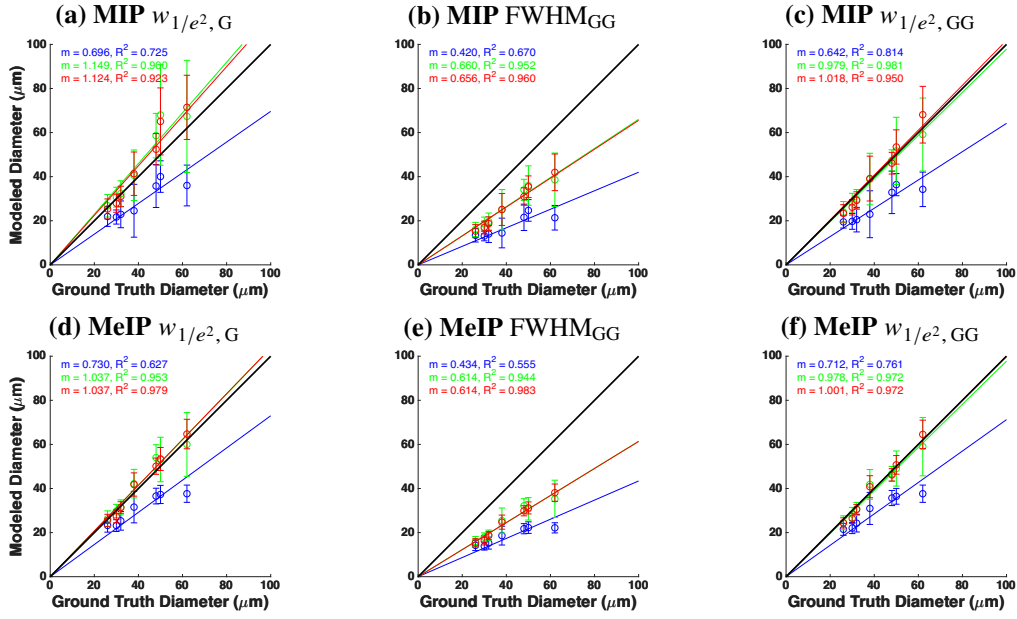


Fig. 4. Modelled versus ground truth vessel diameter using MIP (top row, a–c) and MeIP (bottom row, d–f). Columns correspond to $w_{1/e^2, G}$ (a, d), FWHM_{GG} (b, e), and $w_{1/e^2, GG}$ (c, f). Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. The black line indicates the identity line ($m = 1$).

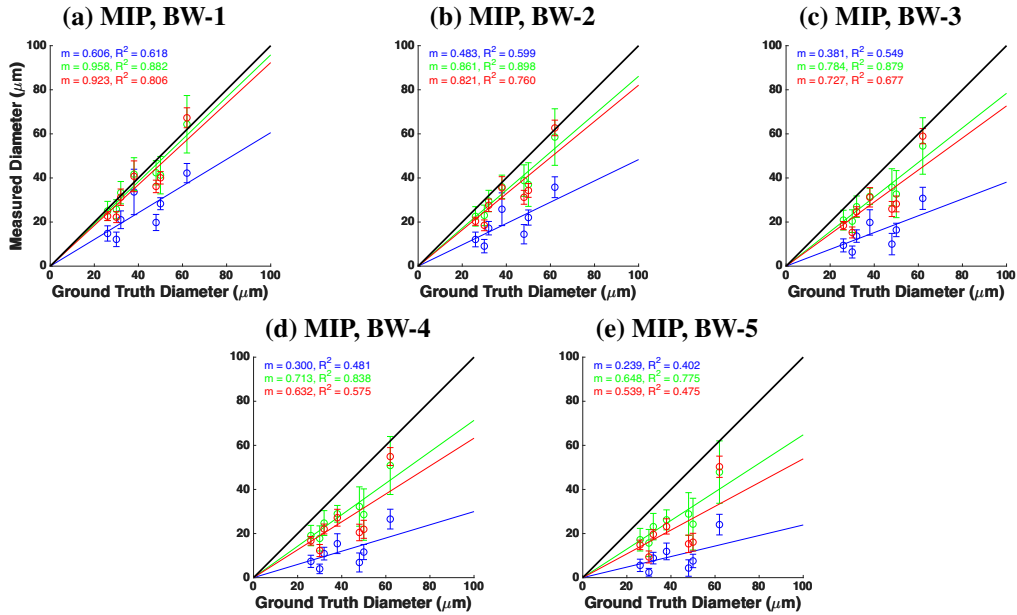


Fig. 5. Vessel diameter measured from binarised masks versus ground truth for MIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

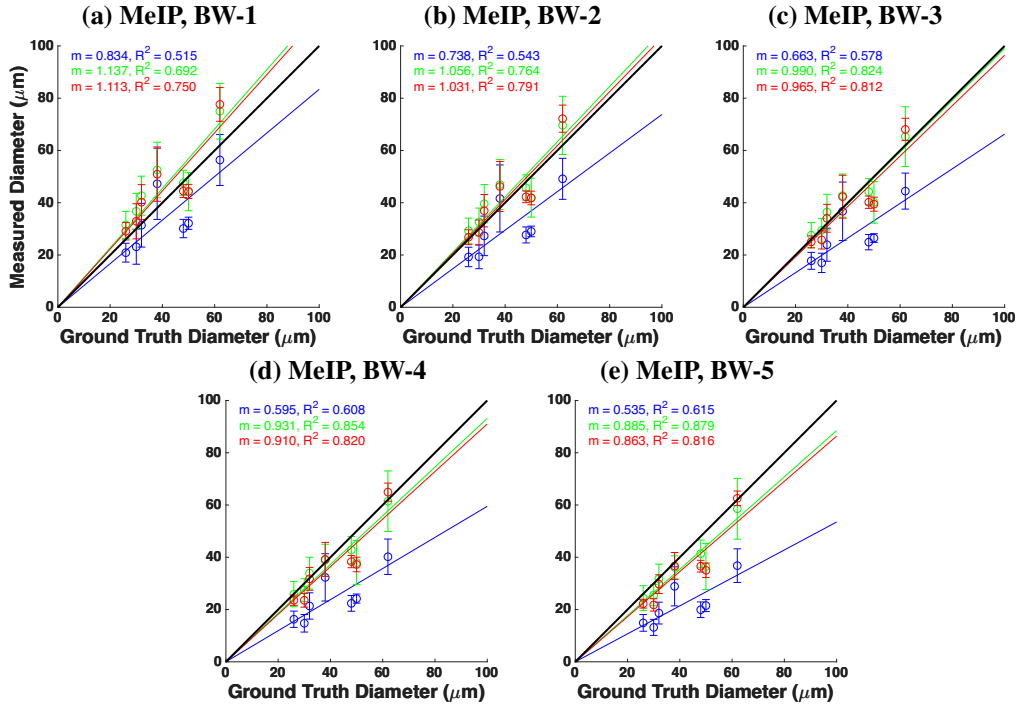


Fig. 6. Vessel diameter measured from binarised masks versus ground truth for MeIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

254 Among the model-based metrics, FWHM_{GG} consistently exhibited the weakest accuracy while
 255 maintaining comparable precision. Overall, model-based methods consistently outperformed
 256 binarisation in both accuracy and precision, and MeIP yielded superior performance compared
 257 to MIP across all methods.

258 3.3. Functional flicker stimulation

259 The model-based diameter time series (Figs. 8a,b) showed increase in diameter post-stimulation
 260 for most of the vessels. The binary mask based diameter time series (BW-3; Fig. 8c) showed a
 261 mixed trend, where most vessels showed a decrease in diameter post-stimulation. Bar plots of the
 262 mean diameter pre- and post-stimulation (Figs. 8d-f) confirmed this pattern. For the Gaussian
 263 model, four of five vessels showed significant diameter increases ($p < 0.01$), and the GG model
 264 showed significant increases in three of five vessels (Table 2). In contrast, the binary mask based
 265 diameter showed significant decreases in four of five vessels.

266 The angiogram intensity time series (Fig. 8g) showed a visible decline in angiogram intensity
 267 during the duration of the DyC-OCT scan across all five vessels. The mean intensity comparison
 268 pre- and post-stimulation (Fig. 8h) confirmed that the decrease was significant for all vessels,
 269 ranging from -14.1% to -19.9% (Table 2). The binary mask based method showed uniformly
 270 positive per-vessel correlations, with an overall correlation of $\rho = 0.523$ ($p < 0.0005$). The
 271 Gaussian and GG models yielded overall correlations of $\rho = -0.101$ and $\rho = -0.098$, respectively.
 272 These results demonstrated that the model-based diameter calculation provided more accurate
 273 and repeatable measurements, and also recovered physiologically meaningful vascular responses
 274 that binary mask based methods misrepresented.

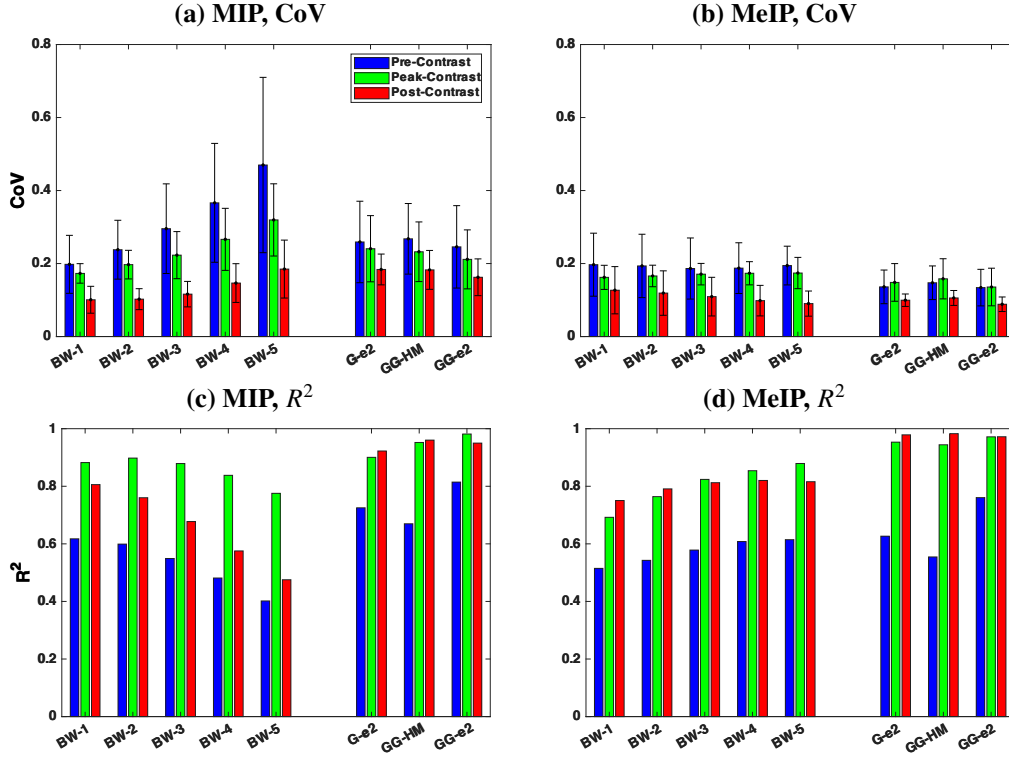


Fig. 7. Summary of accuracy and precision across all diameter measurement methods. (a, b) Coefficient of variation (CoV) and (c, d) coefficient of determination (R^2) for MIP (a, c) and MeIP (b, d), respectively. Bars are grouped by method (BW-1 through BW-5, G- e^2 , GG-HM, GG- e^2) and colour-coded by contrast conditions.

Table 1. Summary of slope (m), coefficient of determination (R^2), and coefficient of variation (CoV) for all diameter measurement methods across pre-contrast, peak contrast, and post-contrast conditions. Methods are grouped as binary mask based and model-based. Bold values indicate slope agreement within 10% of unity ($0.9 \leq m \leq 1.1$), strong linearity ($R^2 \geq 0.90$), or high precision (CoV ≤ 0.10). (confirm values with MATLAB.)

	BW-1		BW-2		BW-3		BW-4		BW-5		$w_{1/e^2, G}$		FWHM _{GG}		$w_{1/e^2, GG}$	
	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP
<i>Slope (m)</i>																
Pre	0.61	0.83	0.48	0.74	0.38	0.66	0.30	0.59	0.24	0.53	0.70	0.73	0.42	0.43	0.64	0.71
Peak	0.96	1.14	0.86	1.06	0.78	0.99	0.71	0.93	0.65	0.88	1.15	1.04	0.66	0.61	0.98	0.98
Post	0.92	1.11	0.82	1.03	0.73	0.97	0.63	0.91	0.54	0.86	1.12	1.04	0.66	0.61	1.02	1.00
<i>Coefficient of determination (R^2)</i>																
Pre	0.58	0.59	0.54	0.62	0.45	0.64	0.34	0.65	0.24	0.65	0.83	0.84	0.79	0.83	0.87	0.88
Peak	0.88	0.78	0.89	0.82	0.86	0.86	0.81	0.88	0.74	0.89	0.87	0.95	0.95	0.95	0.98	0.97
Post	0.76	0.77	0.69	0.79	0.57	0.80	0.45	0.80	0.34	0.78	0.89	0.97	0.95	0.98	0.93	0.97
<i>Coefficient of variation (CoV) (values extracted from bar plots, confirm with MATLAB)</i>																
Pre	0.20	0.20	0.24	0.20	0.29	0.19	0.37	0.19	0.47	0.20	0.26	0.14	0.26	0.15	0.25	0.14
Peak	0.17	0.16	0.20	0.15	0.23	0.17	0.27	0.18	0.32	0.18	0.22	0.16	0.23	0.17	0.21	0.15
Post	0.10	0.12	0.10	0.12	0.11	0.11	0.15	0.10	0.19	0.09	0.19	0.10	0.19	0.11	0.16	0.09

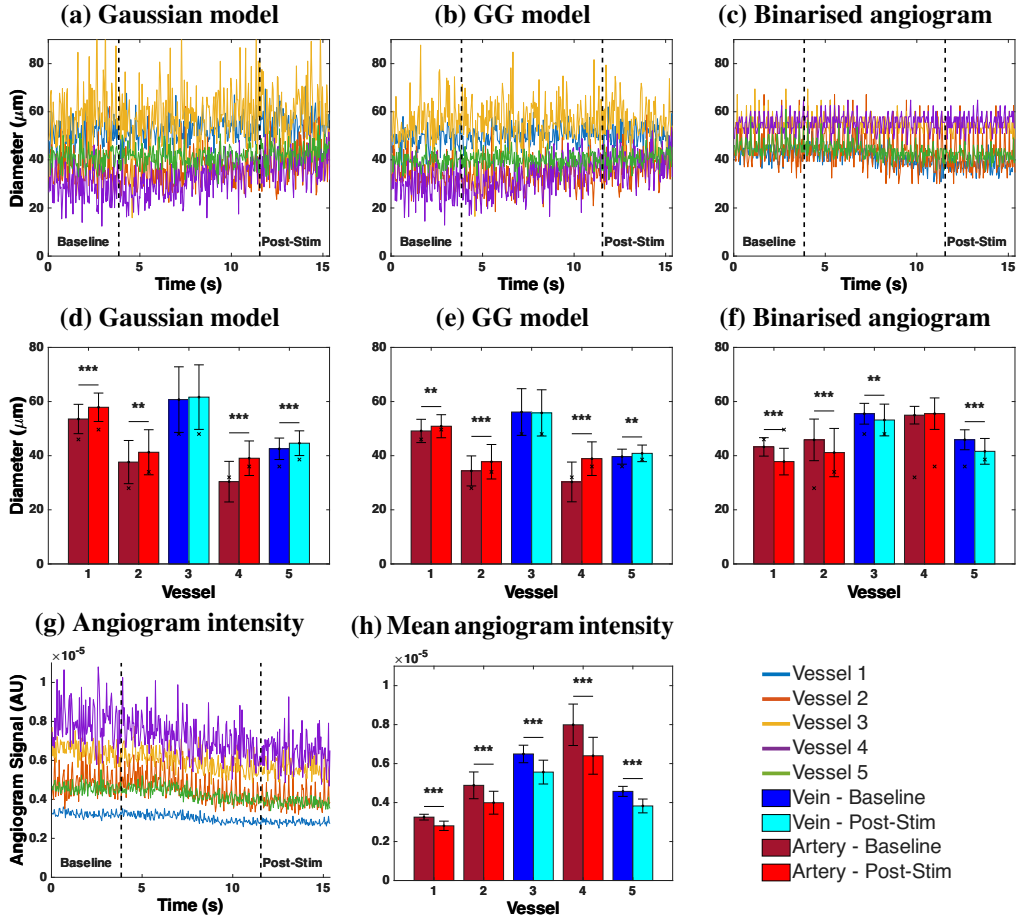


Fig. 8. Functional flicker stimulation results. (a–c) Vessel diameter time series pre- and post-stimulation for (a) Gaussian model, (b) Generalised Gaussian model, and (c) binarised angiogram, across five vessels. Vertical dashed lines indicate stimulation onset and offset. (d–f) Mean diameter per vessel before (dark bars) and after (light bars) stimulation for each method. Veins are shown in blue/cyan and arteries in dark/light red. Significance levels: ** $p < 0.01$, *** $p < 0.001$. (g) Angiogram intensity time series and (h) mean angiogram intensity per vessel pre- and post-stimulation.

Table 2. Flicker stimulation results. Δd : diameter change (%); ρ : Correlation between diameter and angiogram intensity; ΔI : intensity change (%). Statistically significant values are bold (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

	Gaussian		GG		Binarised		Intensity
	Δd (%)	ρ	Δd (%)	ρ	Δd (%)	ρ	
Vessel 1	+9.2***	-0.07	+1.5	0.03	-12.5***	0.59***	-14.1***
Vessel 2	+9.6**	0.23*	+9.9***	0.07	-10.2***	0.80***	-18.2***
Vessel 3	+1.5	-0.28**	-0.6	-0.28**	-4.2**	0.27**	-14.3***
Vessel 4	+28.5***	-0.15	+28.4***	-0.14	+1.0	0.46***	-19.9***
Vessel 5	+4.9***	-0.24*	+3.1**	-0.18	-9.3***	0.52***	-16.3***
Overall ρ		-0.101		-0.098		0.523	

275 4. Discussion

276 4.1. *Model-based vs binary mask based diameter measurement*

277 The model-based and binary mask based diameter methods define vessel boundary in fundamen-
278 tally different ways. Binarisation of the en face maps discards the spatial intensity distribution
279 and reduces each pixel to a binary decision. The measured diameter directly depends on the
280 chosen threshold value and on angiogram intensity, both of which may vary across imaging
281 sessions, contrast conditions, and even between vessels within the same scan. Model-based
282 methods fit a parametric model function to the maximum or mean intensity profiles, and calculate
283 the diameter from the fitted parameters. As the fit captures the shape of the profile rather than
284 the absolute intensity, it is inherently less sensitive to changes in angiogram signal and does not
285 require a varying threshold.

286 This difference explains the trends observed in Fig. 7. Model-based methods maintained
287 strong linearity with the ground truth across all contrast conditions, whereas the binary mask
288 based slopes degraded progressively with increasing threshold. Pre-contrast, the hourglass
289 artefact caused by RBC orientation within vessel lumen suppresses edge signals, causing all
290 methods to underestimate diameter [15]. However, model-based methods still maintained a
291 linear relationship with the ground truth, producing systematic, predictable underestimation.
292 Binary mask methods on the other hand showed much weaker linearity, potentially because the
293 suppressed edge signals did not consistently exceed the binarisation threshold, depending on
294 vessel size and local intensity.

295 The generalised Gaussian (GG) model outperformed the standard Gaussian because the
296 intensity profiles of the vessels are usually not purely Gaussian (Fig. 1b). The standard Gaussian
297 cannot capture relatively flat-topped angiogram intensity profiles and tend to fit a narrower
298 curve, whereas the GG model accommodates through the shape parameter β , which ranges
299 from Gaussian ($\beta = 2$) to increasingly flat-topped profiles ($\beta > 2$). Among the width metrics,
300 FWHM_{GG} consistently showed the weakest accuracy. It could be because the half-maximum
301 determines the vessel boundary where the intensity has already dropped inside the true edge.
302 The $1/e^2$ width captures more of the transition region and so may have provided a more accurate
303 measure of the vessel diameter.

304 4.2. *Maximum vs mean intensity projection*

305 Across nearly all methods and conditions, MeIP outperformed MIP in both accuracy and
306 precision (Table 1). This finding is noteworthy because MIP has been shown to produce higher
307 signal-to-noise ratio (SNR) and contrast in en face OCTA images [12], and is the default in many
308 commercial OCTA systems and analysis pipelines [11, 13]. However, we observed that higher
309 SNR did not necessarily translate to more accurate diameter measurements. The MIP selects the
310 single highest intensity value along each A-scan within the vessel ROI, which makes it inherently
311 sensitive to outliers. MeIP averages all intensity values along each A-scan, which smooths out
312 intensity fluctuations and provides a more average profile. This averaging could be beneficial for
313 vessel diameter measurement, as the goal is to capture the typical vessel boundary rather than
314 extreme signal events.

315 The difference between MIP and MeIP was especially apparent in the binary mask based results
316 (Table 1). For MeIP, the slope declined more gradually and remained above 0.86 even at BW-5
317 post-contrast. Whereas for MIP, the slope decreased steeply with increasing threshold, dropping
318 from 0.92 at BW-1 to 0.54 at BW-5 post-contrast. These findings suggest that MeIP produces a
319 more uniform intensity distribution across the vessel profile, making the binarisation result less
320 sensitive to the exact threshold value. The CoV data (Table 1) reinforced this as the MIP-based
321 binary mask CoV increased substantially with threshold, whereas the MeIP-based CoV remained
322 relatively stable. This behaviour follows from the shape of the two projected profiles. Because the

MIP retains only the brightest pixel along each A-scan, the resulting profile is sharply peaked with steep rises and falls, and the peak intensity could vary considerably between vessels depending on the local signal. Raising the threshold could therefore remove a vessel dependent fraction of the cross-section, and so the measured width scatters more widely about its mean causing increased CoV. Unlike MIP, MeIP averages along each A-scan and produces a broader, smoother profile. As a result, thresholding produces a comparable reduction in measured width across vessels, leading to a largely unchanged CoV despite increasing threshold. For model-based methods, MeIP also showed better performance. The $w_{1/e^2, G}$ using MIP slightly overestimated vessel diameter post-contrast with $m = 1.12$, while MeIP reduced this to $m = 1.04$, a much closer agreement with the ground truth. These results suggest that while MIP may be preferred for visualisation purposes where contrast and SNR are important [12], MeIP should be the preferred projection method for quantitative diameter measurements.

4.3. Best choices under different conditions

Under the conditions where a contrast agent is available and accuracy is the primary goal, the $w_{1/e^2, GG}$ metric using MeIP provides the best results post-contrast. This combination achieved a slope of $m = 1.001$ with $R^2 = 0.97$ and a CoV of 0.09, yielding the best accuracy, linearity, and precision of all tested methods. If only pre-contrast data is available, the same metric still provides the best linearity ($R^2 = 0.88$) among all methods, though the slope of $m = 0.71$ indicates that diameter could be underestimated by approximately 30%. In this case, a correction factor could be applied if absolute diameter values are needed, though relative comparisons between vessels or timepoints would remain valid without correction.

When only MIP data is available, which is the case for many commercial OCTA machines and existing datasets, the $w_{1/e^2, GG}$ still provides good post-contrast accuracy ($m = 1.02$, $R^2 = 0.93$). For binary mask based analysis of MIP data, only the lowest threshold (BW-1) post-contrast achieved a slope near unity ($m = 0.92$), and the slope degraded rapidly with increasing threshold values. This highlights that if binarisation must be used with MIP data, a lower threshold value is essential, though this may come at the cost of increased noise sensitivity.

If computation time is a limiting factor and model fitting is not practical, MeIP with a moderate threshold (e.g. BW-3) provides a reasonable alternative. At post-contrast, BW-3 using MeIP achieved $m = 0.97$ and $R^2 = 0.80$, which is acceptable for many applications. MeIP-based binary mask based analysis showed more stable CoV values across all thresholds compared to MIP, making the results less sensitive to the exact threshold choice. This stability makes MeIP the preferred projection for binary mask based OCTA.

4.4. Influence of angiogram intensity on diameter measurements

The functional flicker stimulation experiment provided a direct demonstration of how changes in angiogram signal intensity can affect diameter measurements based on the method used. During the DyC-OCT acquisition, the angiogram intensity declined significantly across all five vessels (Fig. 8g,h), with decreases ranging from -14.1% to -19.9% (Table 2). This decline is expected during prolonged imaging sessions. Because the angiogram signal scales with blood flow [16], and anaesthesia is known to reduce retinal blood flow [17] and to progressively shift murine ocular physiology and superficial vessel parameters over the course of an imaging session [18], a gradual reduction in retinal perfusion would lower the angiogram signal independently of any functional change in vessel diameter. Simultaneously, the physiological response to flicker stimulation produces vasodilation, which has been well established in retinal physiology [19, 20].

The model-based methods correctly detected this expected vasodilation, with the Gaussian model showing significant diameter increases in four of five vessels (up to $+28.5\%$) and the GG model showing significant increases in three of five vessels (Table 2). The binary mask based method, however, reported significant diameter decreases in four of five vessels, suggesting

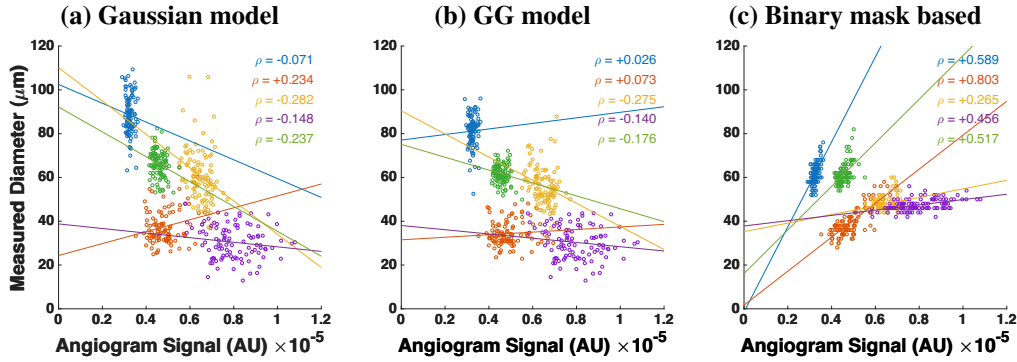


Fig. 9. Measured vessel diameter as a function of angiogram signal intensity during functional flicker stimulation for (a) Gaussian model, (b) Generalised Gaussian model, and (c) binary mask based method. Each colour represents one of five vessels. ρ denotes the Pearson correlation coefficient.

vasoconstriction which is a paradoxical result. The reason behind this behaviour could be understood by observing how each method responds to declining signal intensity (Fig. 9). For the binary mask based method, the threshold is fixed and diameter is determined by the number of pixels that exceed this threshold. When the overall angiogram intensity drops, fewer pixels at the vessel edges exceed the threshold, and the vessel appears to shrink even if the physical vessel diameter has increased. This is confirmed by the strong positive correlation between measured diameter and angiogram intensity for the binary mask based method ($\rho = 0.523$, $p < 0.0005$). In contrast, the model-based methods showed weak overall correlations with intensity ($\rho = -0.101$ for Gaussian, $\rho = -0.098$ for GG), indicating that the fitted diameter was effectively independent of the signal intensity. Among the methods, the GG model also produced the tightest error bars across the five vessels in the flicker experiment (Fig. 8d–f), consistent with its low CoV post-contrast (Table 1), indicating that it is the most repeatable of the metrics tested.

These findings have important implications for longitudinal studies and any experiment where angiogram intensity may vary between or within imaging sessions. In longitudinal mouse eye imaging, the angiogram signal can vary between and within sessions as anaesthesia alters retinal perfusion and ocular physiology [17, 18], with additional factors such as corneal drying and cataract formation further reducing OCT intensity over time [21, 22] if artificial tear drops are not applied during the imaging. If a binary mask based method is used under such conditions, observed changes in vessel diameter may reflect intensity fluctuations rather than true vascular changes. In contrast, model-based methods may be less sensitive to these intensity variations, as they characterise the shape of the intensity profile rather than relying primarily on absolute intensity values to define vessel boundaries.

4.5. Threshold selection

The results presented here used a basic, global threshold for the binarisation of the angiogram data. There are numerous other approaches for binarising the angiogram, which will most likely each behave differently. For example, angiogram signals can be enhanced with vesselness filters [23] and dynamic thresholds can be used to binarise images with varying SNR. When applying a vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is also possible for such enhancement to overcompensate the signal, leading to larger than expected vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance, but similarly risks over- or under-compensating vessels given that threshold level directly affects the measured diameter. Further analysis of these other methods would be required

to determine their efficacy in returning accurate and precise diameter values.

The key observation from the present work is that any thresholding approach, regardless of how the threshold is determined, shares the same fundamental limitation that the measured diameter depends on the absolute signal intensity at the vessel edges. As demonstrated by the flicker stimulation experiment, this intensity dependence can introduce artefactual diameter changes when the angiogram signal varies over time or between sessions. A more adaptive thresholding strategy may reduce variability across different regions of a single image, but it would not eliminate the sensitivity to temporal intensity fluctuations that we observed. The model-based approach avoids this limitation entirely by deriving diameter from the shape of the intensity profile rather than from an intensity cut-off, making it a fundamentally different and more robust alternative for quantitative diameter analysis.

4.6. Literature comparison

Hormel et al. [12] showed that MIP produces en face OCTA images with higher SNR and contrast compared to MeIP, and concluded that MIP should be preferred for OCTA visualisation. Our results are consistent with this for image quality purposes, but reveal a different picture when it comes to quantitative vessel diameter measurement. We found that MeIP consistently yielded more accurate and precise diameter values than MIP across both model-based and binary mask based approaches. This suggests that the projection method best suited for visualisation is not necessarily the best suited for quantification, and that the choice of projection should depend on the intended analysis.

The challenge of standardising OCTA image analysis has been highlighted by Sampson et al. [13] and Untracht et al. [11], both of whom emphasised the need for transparent, reproducible analysis pipelines. Our findings add to this discussion by showing that the choice of diameter measurement method introduces a source of variability that is at least as significant as the choice of binarisation threshold. The model-based approach presented here offers a path toward more standardised vessel diameter quantification, as it reduces the dependence on selected thresholds and angiogram intensity, and provides measurements that are more robust across imaging conditions.

The RBC orientation artefact and its effect on OCTA vessel appearance has been described by Bernucci et al. [15] and Cimalla et al. [6], who used Intralipid contrast to demonstrate that the hourglass pattern in vessel cross-sections is caused by the directional scattering properties of RBCs rather than by true variations in flow. Our work extends this observation by quantifying the degree to which this artefact affects diameter measurements and by proposing model-based fitting as a practical correction strategy. While contrast enhancement with Intralipid is effective for revealing the true vessel diameter, it requires a contrast injection and is not yet applicable in clinical settings. Using a calibrated model-based approach provides a way to partially recover the underestimated diameter from native OCTA data without the need for a contrast agent.

Recent work by Son et al. [24] used functional OCT to image flicker-evoked vasodilation in the mouse retina and reported that vasodilation is anisotropic, with larger percent changes in the axial direction ($\sim 18\%$) than the lateral direction ($\sim 11\%$). The authors did not describe how vessel diameters were quantified from their B-scans. It is worth considering whether the lateral diameter measurements may have been affected by the intensity-dependent artefacts observed in the present work. In the lateral direction, the RBC orientation artefact suppresses signal at vessel edges, and depending on how their vessel boundary was defined, this artefact could lead to an underestimation of the lateral diameter. If this were the case, the apparent difference between axial and lateral vasodilation could be influenced by the measurement methodology in addition to the biomechanical anisotropy discussed by the authors. It would be interesting to apply the model-based approach presented here to such data to determine whether their reported anisotropy changes when the lateral diameter is quantified differently.

The magnitude of vasodilation detected by our model-based methods ranged from +1.5% to

452 +28.5% across the five vessels measured (Table 2). The arteries (Vessels 1, 2, and 4) showed
 453 the largest diameter increases, with the Gaussian model detecting changes of +9.2%, +9.6%,
 454 and +28.5%, respectively, while the veins (Vessels 3 and 5) showed smaller diameter increases
 455 of +1.5% and +4.9%. Table 3 summarises the stimulation protocols and vascular responses
 456 reported by previous flicker stimulation studies alongside our own. The listed human studies
 457 span several measurement techniques, from fundus photography [25] through retinal vessel
 458 analysers [19, 26, 27] to Doppler OCT [28] and OCTA [29]. Despite these methodological
 459 differences, the reported human arterial responses cluster between +3% and +8%, and venous
 460 responses between +2% and +10%. Notably, these studies report arterial and venous responses
 461 of broadly comparable magnitude, and in some cases the venous response is the larger of the two,
 462 for example +9.8% venous versus +8.0% arterial in Aschinger et al. [28]. This contrasts with
 463 our measurements, in which the arterial response was consistently and substantially larger than
 464 the venous response. Whether this asymmetry reflects a genuine feature of the murine flicker
 465 response, the small number of vessels sampled, or a greater sensitivity of the model-based fit
 466 to the smooth-muscle-driven dilation of arteries remains to be established. Polak et al. [26]
 467 additionally showed that the diameter response is relatively stable across flicker frequencies from
 468 4 to 40 Hz, suggesting that frequency differences between studies are unlikely to explain the
 469 variation. The magnitudes we observed are generally larger than those reported in the human
 470 studies, though direct comparison is difficult given the differences in experimental conditions.
 471 Species, anaesthesia, stimulation frequency and wavelength, stimulus power and delivery method,
 472 stimulation duration, dark adaptation period, and imaging modality all vary across the reported
 473 studies, and each of these factors could influence the measured vascular response. Among the
 474 studies listed, Son et al. [24] used the most similar setup to ours. They used the same mouse
 475 strain (C57BL/6J), the same anaesthetic (ketamine/xylazine), a comparable flicker stimulus
 476 (10 Hz, 504 nm), however a different stimulation period of 5 s. A systematic comparison of these
 477 parameters could help determine which factors are most influential to the results observed in
 478 both studies. Importantly, regardless of the absolute magnitude, the fact that we were able to
 479 detect vasodilation at all depended on the measurement method, because binary mask based
 480 analysis not only failed to detect vasodilation but reported vasoconstriction which would have led
 481 to fundamentally incorrect physiological conclusions.

482 Most other studies in mice have measured flicker-evoked retinal vessel diameter changes using
 483 dynamic vessel analysis (DVA), which is a fundus camera-based technique adapted from human
 484 retinal vessel analysers. Albanna et al. [30] demonstrated the feasibility of DVA in the mouse
 485 eye using 12.5 Hz flicker at 530 nm but found that quantitative arterial recordings were possible
 486 only in a minority of animals, while venous dilation was mostly below 1.5% post-stimulation.
 487 Another study by Neumaier et al. [31] used the same setup in wildtype and Cav2.3-deficient mice
 488 with overnight dark adaptation and reported median arterial and venous diameter changes of only
 489 +1.3% and +1.2%, respectively, in the wildtype group. These values are substantially smaller
 490 than those we observed, which could be due to the limited spatial resolution and signal-to-noise
 491 ratio of DVA when applied to the smaller mouse eye rather than being a genuine physiological
 492 response. The much larger responses we observed may therefore partly result from the higher
 493 lateral resolution of OCT combined with the model-based fitting approach, which captures
 494 vessel boundary shifts that could fall below the detection limit of DVA. In addition to these
 495 diameter-based studies, Rai et al. [32] recently reported that 12 Hz flicker stimulation significantly
 496 increased both arterial and venous retinal blood flow in light-adapted C57BL/6J mice, confirming
 497 that the mouse retina exhibits a robust neurovascular coupling response. Since blood flow is
 498 proportional to the product of vessel cross-sectional area and flow velocity, their findings are
 499 consistent with the vasodilation we observed, although a direct quantitative comparison between
 500 flow and diameter changes is not straightforward.

Table 3. Comparison of flicker stimulation protocols and reported diameter changes across studies. H = human, M = mouse. DA = dark adaptation. Lat. = lateral, ax. = axial.

Study	Stimulation	Stimulation	Stimulation	DA	Diameter Change (%)	
	Freq. (Hz)	Light Source	Duration (s)	(h)	Artery	Vein
Formaz [25] (H)	10	Xenon arc, 5.4 log Td [§]	60	—	+4.2	+2.7
Polak [26] (H)	8	< 550 nm, ~ 150 $\mu\text{W}/\text{cm}^{-2}\dagger$	60	—	+3.5	+1.9
Garhofer [19] (H)	8	< 550 nm, 300 $\mu\text{W}/\text{cm}^{-2}\dagger$	60	—	+3.8	+2.8
Nagel [27] (H)	12.5	530 – 600 nm, ~ 196 $\mu\text{W}/\text{cm}^{-2}\dagger$	20	—	+6.9	+6.5
Aschinger [28] (H)	12.5	530 – 600 nm, ~ 196 $\mu\text{W}/\text{cm}^{-2}\dagger$	120	—	+8.0	+9.8
Kallab [29] (H)	7	White, ~ 450 lux	*	—	+3.7	+4.5
Albanna [30] (M)	12.5	530 nm, 30 lux	20	12	— [#]	< 1.5 [#]
Neumaier [31] (M)	12.5	530 nm, 30 lux	20	12	+1.3	+1.2
Son [24] (M)	10	504 nm, ~ 460 lux	5	1 – 2	lat. +10.8, ax. +18.2	lat. +8.9, ax. +14.3
Present study (M)	10	502 nm, 26.6 $\mu\text{W}/\text{cm}^2$ [‡]	Cont.	2	+9.2 to +28.5	+1.5 to +4.9

[§]Diameter measured from fundus photographs taken post-stimulation; retinal luminance in trolands.

[#]DVA feasibility study; arterial recordings were not reliably obtainable in the mouse eye.

*Flicker applied during OCTA scan acquisition. [†]Retinal irradiance. [‡]Power at cornea.

4.7. Future direction

A future direction of this work is to move toward fully automated segmentation and calculation of vessel diameters from en face angiogram maps. While ROIs were manually selected in this work, it is also possible to automatically segment angiographic data using traditional skeletonisation methods, which are commonly used for automated OCTA analysis. From a skeletonised image, vessel angle can be computed for further extraction of vascular cross-sections perpendicular to the vessel axis. These cross-sections could then be processed using the model-based methods described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-automated approach shown here for 2D OCTA data. This type of model-based approach is particularly important for comparing 3D volumes acquired at different time points in the mouse eye, as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we have shown in this work, even low levels of intensity loss occurring over a short time window can significantly affect the measured vessel diameter when using a binary mask based approach, but not when using the models presented here.

Additionally, this work focused on the larger vessels of the superficial vascular plexus. Extending the model-based approach to the intermediate and deep capillary plexuses, where vessels are smaller and signal levels are lower, would be valuable for studying microvascular changes in diseases. For very small vessels approaching the lateral resolution limit of the imaging system, the measured intensity profile is a convolution of the true vessel profile with the point spread function (PSF) of the incident beam on the retina. Since the lateral resolution of OCT is typically worse than the axial resolution and is determined by the beam optics rather than the source bandwidth, this convolution becomes increasingly significant for smaller vessels. If the PSF were known, a deconvolution approach could be used to recover the true vessel profile before model fitting, potentially improving accuracy for capillary-scale vessels. However, characterising the retinal PSF in vivo remains challenging, as it depends on the ocular optics of each individual eye and may vary with retinal depth and eccentricity.

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